

University of Cape Town

Bayesian Computational Methods

Assignment

Bayesian Analysis Assignment

Author:
Isabella Lethbridge
Tayyeb Abrahams

Student Number:
LTHISA001
ABRMOE030

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Introduction

Question a

Each of the n birds is caught exactly once. The capture day of bird i is denoted by t_i and is treated as fixed. The start of moult for bird i, T_i, is an unobserved random variable, where

$$T_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2), \qquad f_T(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right), \qquad i = 1, \ldots, n. \tag{1}$$

Every bird moults at the same constant rate, with moult lasting τ days for all birds. The moult index of bird i therefore increases linearly from 0 at time T_i to 1 at time $T_i + \tau$, so that on its capture day

$$y_i = \begin{cases} 0, & T_i > t_i, \\ \dfrac{t_i - T_i}{\tau}, & t_i - \tau \le T_i \le t_i, \\ 1, & T_i < t_i - \tau. \end{cases} \tag{2}$$

In the Type 1 model the moult index is not observed. Only the status $s_i \in \{1, 2, 3\}$ of each bird is observed, where $s_i = 3$ if $y_i = 0$, $s_i = 2$ if $0 < y_i < 1$ and $s_i = 1$ if $y_i = 1$. From Equation 2, each status corresponds to an interval of values of T_i. Let $Z_i = \frac{T_i - \mu}{\sigma} \sim \mathrm{N}(0, 1)$ and let Φ denote the standard normal CDF.

For $s_i = 3$, bird i has not yet started moult

$$P(t_i) = Pr(T_i > t_i) = Pr\left(Z_i > \frac{t_i - \mu}{\sigma}\right) = 1 - \Phi\left(\frac{t_i - \mu}{\sigma}\right). \tag{3}$$

For $s_i = 2$, bird i is currently moulting

$$Q(t_i) = Pr(t_i - \tau \le T_i \le t_i) = Pr\left(\frac{t_i - \tau - \mu}{\sigma} \le Z_i \le \frac{t_i - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \mu}{\sigma}\right) - \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{4}$$

For $s_i = 1$, bird i has finished moulting

$$R(t_i) = Pr(T_i \le t_i - \tau) = Pr\left(Z_i \le \frac{t_i - \tau - \mu}{\sigma}\right) = \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right). \tag{5}$$

Every bird is in exactly one of the three states on its capture day, since T_i must fall in exactly one of the intervals $T_i > t_i$, $t_i - \tau \le T_i \le t_i$ or $T_i < t_i - \tau$. The intervals overlap only at the single point $T_i = t_i - \tau$, which has probability zero because T_i is continuous. Thus,

$$P(t_i) + Q(t_i) + R(t_i) = 1. \tag{6}$$

Given t_i, the status s_i is a single categorical draw with probabilities $P(t_i), Q(t_i)$ and $R(t_i)$, given in Equation 3, Equation 4 and Equation 5. The conditional probability of s_i can be written as

$$Pr(s_i \mid t_i, \mu, \sigma^2, \tau) = P(t_i)^{I_{\{s_i=3\}}}\, Q(t_i)^{I_{\{s_i=2\}}}\, R(t_i)^{I_{\{s_i=1\}}}, \tag{7}$$

where $I_{\{*\}} = 1$ if condition $*$ is met and 0 otherwise. The start dates are independent and each bird is caught only once, so the statuses are independent given $\boldsymbol{t}$ and the parameters. Thus, the likelihood is the product of Equation 7 over all n birds

$$\begin{aligned} L(\boldsymbol{s}, \boldsymbol{t}, \mu, \sigma^2, \tau) &= \prod_{i=1}^{n} P(t_i)^{I_{\{s_i=3\}}}\, Q(t_i)^{I_{\{s_i=2\}}}\, R(t_i)^{I_{\{s_i=1\}}} \\ &= \prod_{i=1}^{n} \left[1 - \Phi\left(\frac{t_i - \mu}{\sigma}\right)\right]^{I_{\{s_i=3\}}} \left[\Phi\left(\frac{t_i - \mu}{\sigma}\right) - \Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right)\right]^{I_{\{s_i=2\}}} \\ &\quad \times \left[\Phi\left(\frac{t_i - \tau - \mu}{\sigma}\right)\right]^{I_{\{s_i=1\}}}. \end{aligned} \tag{8}$$

A mixture model is appropriate because, on any capture day t, the population consists of birds in three different states: birds that have not yet started to moult, birds that are undergoing moult and birds that have completed moult. These states occur in proportions $P(t)$, $Q(t)$ and $R(t)$, which depend on t through the parameters μ, σ^2 and τ. Thus, the moult index has a mixture distribution with point masses at 0 and 1 with weights $P(t)$ and $R(t)$, respectively, as well as a continuous component on $(0, 1)$ with weight $Q(t)$. The continuous component is $f_y(y) = \tau f_T(t - y\tau)$, the density of the moult index of a moulting bird, and the substitution $u = t - y\tau$ shows that its total mass is

$$\int_0^1 \tau f_T(t - y\tau)\, dy = \int_{t-\tau}^{t} f_T(u)\, du = Q(t). \tag{9}$$

In the Type 1 model, the moult index is not observed. We only observe the state that bird i belongs to. Thus, each bird contributes the weight of the state it belongs to, $P(t_i)$, $Q(t_i)$ or $R(t_i)$, to Equation 8.

Question b

In the Type 2 model, the moult scores (y_i) of birds that are currently moulting are observed. A Gibbs sampler is developed using data augmentation, by treating the unobserved start dates of the birds that have not yet started moulting and of those that have finished moulting as latent variables. The derivations of the Type 2 likelihood and of the conditional distributions are given in Appendix A.

Definitions

The birds are grouped according to their states. Let $\mathcal{F} = \{i : s_i = 1\}$ denote the set of birds that have already moulted, $\mathcal{M} = \{i : s_i = 2\}$ denote the set of birds that are undergoing moult and $\mathcal{N} = \{i : s_i = 3\}$ denote the set of birds that have not yet started to moult. Let $m = |\mathcal{M}|$ be the number of birds currently moulting. For the `sanderlings` data, $n = 164$, $|\mathcal{F}| = 13$, $m = 66$ and $|\mathcal{N}| = 85$.

For a bird undergoing moult, the linear moult assumption gives $T_i = t_i - y_i\tau$ and the probability density function of its observed moult score is

$$f_y(y_i) = \tau f_T(t_i - y_i\tau), \qquad 0 < y_i < 1. \tag{10}$$

The start dates T_i for $i \in \mathcal{F} \cup \mathcal{N}$ are treated as latent variables and $T_i = t_i - y_i\tau$ for $i \in \mathcal{M}$. The prior distributions are

$$\mu \sim \text{Uniform}(100, 160), \quad \sigma^2 \sim \text{IG}(1, 1), \quad \tau \sim \text{Uniform}(50, 180). \tag{11}$$

The following quantities are used in the conditional distributions below

$$\begin{aligned} \bar{T} &= \frac{1}{n}\sum_{i=1}^{n} T_i, \quad & S &= \sum_{i=1}^{n}(T_i - \mu)^2, & h &= \min_{i\in\mathcal{F}}(t_i - T_i), \\ \beta &= \frac{\sum_{i\in\mathcal{M}} y_i^2}{2\sigma^2}, & \gamma &= \frac{\sum_{i\in\mathcal{M}} y_i(t_i - \mu)}{\sigma^2}, & \bar{t}_y &= \frac{\sum_{i\in\mathcal{M}} y_i t_i}{\sum_{i\in\mathcal{M}} y_i}, \end{aligned} \tag{12}$$

where $T_i = t_i - y_i\tau$ is used for $i \in \mathcal{M}$ and $V = [50, 180] \cap [0, h]$.

Conditional distributions

The latent start dates follow the same truncated normal distributions as in the Type 1 Gibbs sampler

$$\begin{aligned} T_i \mid \mu, \sigma^2, \tau, s_i = 1, \qquad & t_i \sim \text{N}(\mu, \sigma^2)\, I_{\{T_i \le t_i - \tau\}}, \quad i \in \mathcal{F}, \\ T_i \mid \mu, \sigma^2, \tau, s_i = 3, \qquad & t_i \sim \text{N}(\mu, \sigma^2)\, I_{\{T_i > t_i\}}, \qquad i \in \mathcal{N}. \end{aligned} \tag{13}$$

Unlike in the Type 1 model, the start dates of the moulting birds are not latent, since they are determined by τ through $T_i = t_i - y_i\tau$. The full conditional distributions of μ and σ^2 are

$$\mu \mid \sigma^2, \tau, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T} \sim \text{N}\left(\bar{T}, \frac{\sigma^2}{n}\right) I_{\{100 \le \mu \le 160\}}, \tag{14}$$

$$\sigma^2 \mid \mu, \tau, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T} \sim \text{IG}\left(\frac{n}{2} + 1,\ 1 + \frac{S}{2}\right). \tag{15}$$

The full conditional distribution of τ is a truncated modified half-normal (MHN) distribution (Sun et al., 2021)

$$\tau \mid \mu, \sigma^2, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T} \sim \text{MHN}(m+1,\ \beta,\ \gamma)\, I_{\{50 \le \tau \le \min(180,h)\}}. \tag{16}$$

In the Type 1 model, τ appears only in the indicators and has a uniform conditional distribution on V. In the Type 2 model, the observed moult scores result in a non-uniformly shaped conditional distribution of τ. The moulting birds no longer have latent start dates, so there is no lower bound l from the data. Since the start dates of the finished birds are drawn such that $T_i \le t_i - \tau$, we have $h \ge \tau \ge 50$ and thus V is never empty.

Sampling from Equation 16 is simplified by a further data augmentation step based on Lemma 5 of Sun et al. (2021), the form of which depends on the sign of γ. Since $y_i > 0$ and $\sigma^2 > 0$, $\gamma \ge 0$ if and only if $\mu \le \bar{t}_y$. For the `sanderlings` data, $\bar{t}_y = 164.4$, which exceeds the upper prior bound of μ. Thus, $\gamma > 0$ at every iteration and a latent variable W is introduced such that

$$\begin{aligned} W \mid \tau &\sim \text{Poisson}(\gamma\tau), \\ \tau^2 \mid W &\sim \text{Gamma}\left(\frac{m+1+W}{2},\ \beta\right) I_{\{50 \le \tau \le \min(180,h)\}}. \end{aligned} \tag{17}$$

The gamma distribution is parameterised by shape and rate. The truncated gamma distribution is sampled exactly using the inverse CDF method and τ is obtained as the positive square root of the draw. The latent variable W is redrawn given the current τ at every iteration and then discarded, so the conditional distributions of the other parameters are unchanged.

The Gibbs sampling algorithm

At least three chains are run, each from random starting values. At each step, the most recently sampled values of all other variables are used.

1. Initialise. Draw starting values $\mu \sim \text{Uniform}(100, 160)$, $\sigma^2 \sim \text{IG}(1, 1)$ and $\tau \sim \text{Uniform}(50, 180)$. Then draw T_i for $i \in \mathcal{F} \cup \mathcal{N}$ from Equation 13.
2. Repeat for K iterations:
   a. Draw T_i for $i \in \mathcal{F} \cup \mathcal{N}$ from Equation 13.
   b. Set $T_i = t_i - y_i\tau$ for $i \in \mathcal{M}$.
   c. Draw μ from Equation 14.
   d. Draw σ^2 from Equation 15.
   e. Compute h, β and γ from Equation 12.
   f. Draw $W \sim \text{Poisson}(\gamma\tau)$ and then τ^2 from the truncated gamma distribution in Equation 17. Set τ to the positive square root.
3. Output. Discard the burn-in iterations and keep the draws of (μ, σ^2, τ). The latent variables T_i and W are not kept.

Question c

A random walk Metropolis-Hastings algorithm is used to sample from the posterior distribution of the Type 3 model, with the likelihood given in the assignment and the priors in Equation 11. The random walk steps are drawn from a multivariate normal distribution, which has support over the entire real line, whereas the parameters do not. The prior restricts the mean start date μ to the interval $(100, 160)$ and the duration of moult τ to $(50, 180)$. The variance σ^2 is constrained to be positive. Thus, each parameter is mapped to the real line using the transformations

$$\phi_1 = \text{logit}\left(\frac{\mu - 100}{60}\right), \qquad \phi_2 = \log \sigma^2, \qquad \phi_3 = \text{logit}\left(\frac{\tau - 50}{130}\right). \tag{18}$$

These transformations each have support over the entire real line, so we may perform our random walk over the transformed space $\boldsymbol{\phi} = (\phi_1, \phi_2, \phi_3)$, with proposals $\boldsymbol{\phi}^* = \boldsymbol{\phi} + \boldsymbol{e}$. However, since we sample $\boldsymbol{\phi}$ rather than our original parameters $\boldsymbol{\theta} = (\mu, \sigma^2, \tau)$, we must include the Jacobian of the transformation in the log posterior to ensure that the Metropolis-Hastings algorithm converges to the correct posterior. With $p_j = \text{logit}^{-1}(\phi_j)$, the log posterior becomes

$$\begin{aligned} \log p(\boldsymbol{\phi}) &= \log L(\mu, \sigma^2, \tau; \boldsymbol{y}) + \log \pi(\boldsymbol{\theta}) + \log |J(\boldsymbol{\phi})|, \\ \log |J(\boldsymbol{\phi})| &= \log\{60\, p_1(1 - p_1)\} + \phi_2 + \log\{130\, p_3(1 - p_3)\}. \end{aligned} \tag{19}$$

Since the proposal is symmetric, $\boldsymbol{\phi}^*$ is accepted with probability $\min\{1, \exp[\log p(\boldsymbol{\phi}^*) - \log p(\boldsymbol{\phi})]\}$. Since we are unsure how our parameters are correlated, the steps of our random walk are initially drawn from

$$\boldsymbol{e} \sim \text{N}(\boldsymbol{0},\, c\, \mathbf{I}_3), \qquad c = 0.5^2,$$

where $\mathbf{I}_3$ is the 3×3 identity matrix. However, we would like to tune our random walk to have a more appropriate covariance matrix. To achieve this, we use the first 3 000 iterations of each chain as a burn-in, and estimate the covariance matrix Σ of our transformed parameters from the last 1 500 of these iterations. Each of our retained draws then has its step drawn from

$$\boldsymbol{e} \sim \text{N}\left(\boldsymbol{0},\, \frac{2.38^2}{3}\, \Sigma\right),$$

where $2.38^2/d$, with $d = 3$ parameters, is the standard scaling for random walk proposals (Roberts et al., 1997). Since Σ is fixed after the burn-in, the Metropolis-Hastings algorithm then proceeds as per usual, with the posterior as its stationary distribution. If the estimated covariance matrix is degenerate because the chain barely moved, we restart the chain from a new starting value.

We run 50 000 retained iterations on three chains (seed 2026), which are each randomly initialised. The parameters μ and τ are drawn from uniform distributions over their prior

ranges and σ^2 is drawn from IG$(1, 1)$. The initial values for our chains, as well as the acceptance rates of our sampler, are given in Table 1.

Table 1: Random starting values and acceptance rates (burn-in and retained iterations) of the three Metropolis-Hastings chains.

Chain	$\mu^{(0)}$	$\sigma^{2(0)}$	$\tau^{(0)}$	Burn-in acceptance	Acceptance rate
1	141.92	1.65	87.14	0.21	0.27
2	115.56	0.76	114.27	0.21	0.34
3	109.01	2.55	74.15	0.20	0.23

The acceptance rates of the retained iterations, between 0.23 and 0.34, are close to the optimal rate of 0.234 (Roberts et al., 1997) and within the efficient range of approximately 0.15 to 0.5 (Roberts and Rosenthal, 2001). This suggests that the step sizes of our random walk are neither too small (which would result in high acceptance rates) nor too large (which would result in low acceptance rates by jumping into sparse areas of the density), and that our covariance matrix was adequately tuned in the burn-in draws.

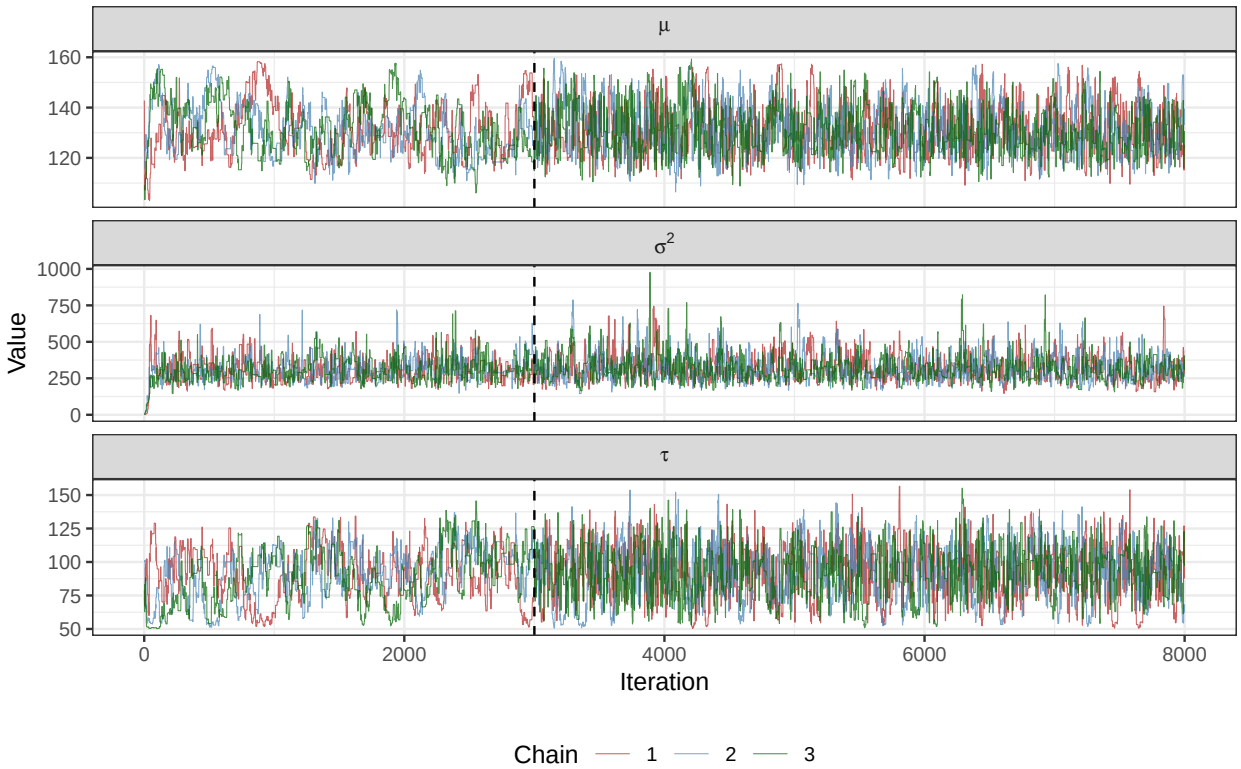


Figure 1: Trace plots of the three Metropolis-Hastings chains for the first 8 000 iterations. The end of the burn-in is indicated by the dashed line.

To assess the convergence of our chains, we inspect the trace plots in Figure 1 and compute

the Gelman-Rubin statistic $\hat{R}$ (Gelman and Rubin, 1992). The trace plots show that the three chains reach the same region within a few hundred iterations and are stationary thereafter, so a burn-in of 3 000 iterations is sufficient. The value of $\hat{R}$ is 1.00 for all parameters (Table 2), so we conclude that our chains have converged. We should note, however, that the effective sample size of our sampler is much smaller than the number of retained draws because of autocorrelation in the draws. The ACF plots in Figure 3 (Appendix B) show that it takes approximately 50 iterations before the draws become uncorrelated. It is therefore unsurprising that the effective sample sizes are only between 8 191 and 9 340, compared to 150 000 retained draws.

Finally, the posterior summaries and 95% credible intervals for our parameters are given in Table 2. We can more directly inspect the marginal densities and bivariate distributions in Figure 4 (Appendix B).

Table 2: Posterior summaries for the Type 3 model obtained using the tuned random walk Metropolis-Hastings algorithm (3 chains, 50 000 retained draws each).

	Mean	SD	2.5%	Median	97.5%	ESS	$\hat{R}$
μ	130.70	9.54	114.82	129.63	151.54	8191	1.00
σ^2	319.19	88.67	193.13	303.90	532.89	9340	1.00
τ	92.68	19.13	55.85	93.53	128.55	8734	1.00
σ	17.71	2.36	13.90	17.43	23.08	9613	1.00

We should note that the credible interval for μ is quite wide and spans a large part of the interval defined by the prior. Since 55 of the 66 moulting birds were caught on a single day (day 127), the data cannot easily separate an early start with a long moult from a late start with a short moult. The posteriors of μ and τ are also cut off at the prior bounds of 160 and 50, respectively, so the prior partly determines the posterior. The credible interval for τ, however, is notably narrower than its prior. We note that while μ and σ^2 appear to be broadly uncorrelated ($\rho = 0.11$), τ is weakly correlated with σ^2 ($\rho = 0.31$) and strongly negatively correlated with μ ($\rho = -0.83$). This is expected, since $T_i = t_i - y_i\tau$ implies that a longer duration gives earlier start dates, and hence a smaller μ, and that the spread in start dates scales with τ.

Question d

We use Stan (Carpenter et al., 2017; Stan Development Team, 2025) to sample from the posterior distribution of the Type 3 model. Stan uses the No-U-Turn Sampler (NUTS), an adaptive form of Hamiltonian Monte Carlo (Hoffman and Gelman, 2014). NUTS uses the gradient of the log posterior to propose new points. Thus, the draws are less autocorrelated than those of our Metropolis-Hastings sampler.

Our Stan program consists of the following blocks. The `data` block contains the capture days t_i and moult scores y_i of the $m = 66$ moulting birds. The `parameters` block declares

μ, σ^2 and τ, restricted to $(100, 160)$, $(0, \infty)$ and $(50, 180)$, respectively. Stan maps these parameters to the real line using the same transformations as in Equation 18 and includes the Jacobian automatically. The `model` block contains the priors in Equation 11 and the Type 3 log-likelihood, where $\log Q(t_i)$ is computed using `log_diff_exp` for numerical stability.

To allow for comparison with Question c, we use the same number of chains, iterations and burn-in as for the Metropolis-Hastings sampler. We run three chains of 53 000 iterations each (seed 2026). The first 3 000 iterations of each chain are warmup iterations, which are used to tune the sampler and are then discarded. All other settings are Stan's defaults, with a target acceptance rate of 0.8, a maximum tree depth of 10 and random starting values drawn from Uniform$(-2, 2)$ on the transformed scale. The trace plots in Figure 5 (Appendix B) show that the chains mix well from the start and $\hat{R}$ is 1.00 for all parameters, with 0 divergent transitions. The posterior summaries are given in Table 3 and the joint posterior distribution is shown in Figure 6 (Appendix B).

Table 3: Posterior summaries for the Type 3 model obtained using Stan (NUTS), based on 3 chains of 50 000 post-warmup draws.

	Mean	SD	2.5%	Median	97.5%	ESS	$\hat{R}$
μ	130.82	9.51	114.99	129.77	151.39	41136	1.00
σ^2	318.91	86.86	194.68	304.22	526.74	57034	1.00
τ	92.42	19.21	55.09	93.28	128.31	44290	1.00
σ	17.71	2.32	13.95	17.44	22.95	59817	1.00

Question e

We compare the posterior distributions obtained from the Metropolis-Hastings (MH) sampler in Question c and the Stan sampler in Question d; the top row of Figure 2 overlays the marginal densities and the bottom row are the corresponding quantile-quantile plot. There is strong visual agreement in the posterior marginal distributions obtained by both samplers and we conclude that the MH sampler has been implemented correctly and converged to the posterior distribution.

We may also compare our estimates obtained by each sampler; for each parameter, we report the difference in the posterior means (MH minus Stan), the pooled variance s_p^2 of the two sets of draws, and the p-value of a two-sample t-test of the null hypothesis that the posterior means are equal. Since the draws of our samplers are autocorrelated, the standard error utilizes the effective sample size as opposed to the number of iterations $\sqrt{s_p^2 \left(1/\mathrm{ESS}_{\mathrm{MH}} + 1/\mathrm{ESS}_{\mathrm{Stan}}\right)}$, the resulting t-distribution has $\mathrm{ESS}_{\mathrm{MH}} + \mathrm{ESS}_{\mathrm{Stan}} - 2$ degrees of freedom. The table also gives the differences in the 2.5%, 50% and 97.5% posterior quantiles and the Kolmogorov-Smirnov distance D between the two sets of draws.

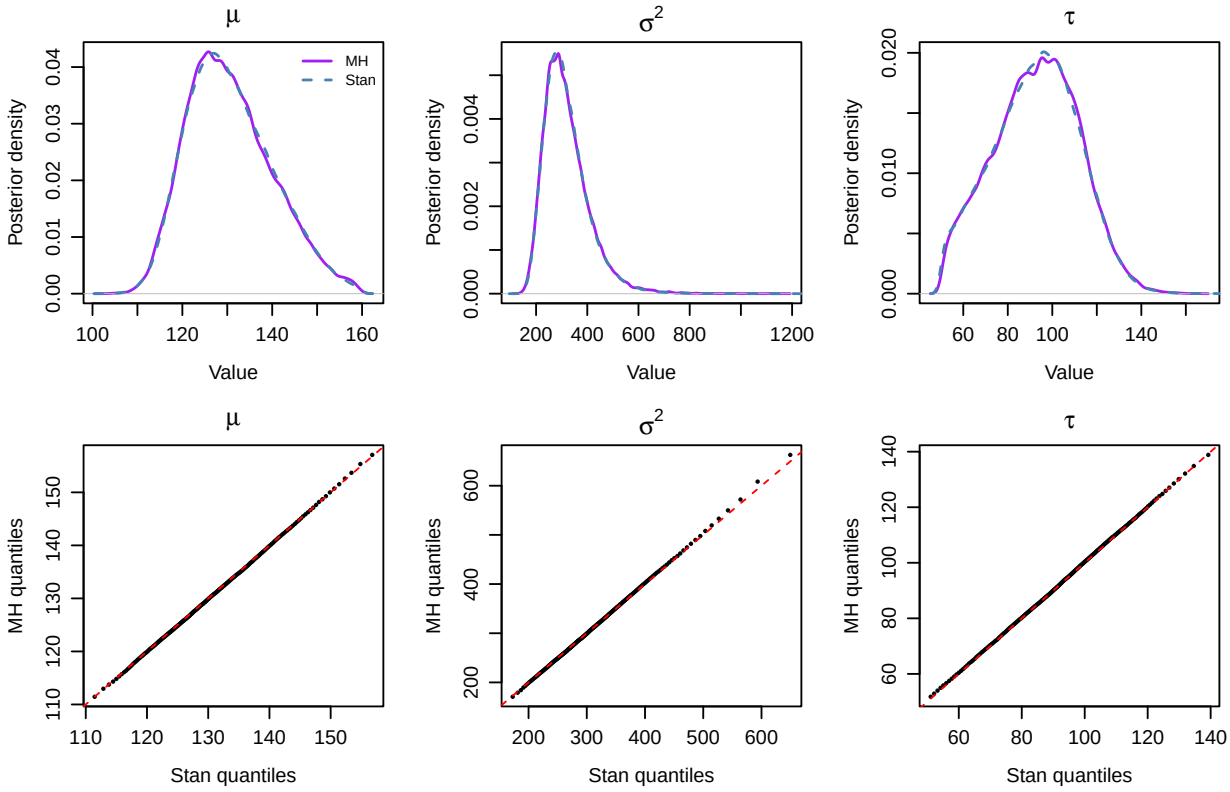


Figure 2: Comparison of the Metropolis-Hastings (MH) and Stan (NUTS) posteriors. Top row: marginal posterior densities. Bottom row: quantile-quantile plots of the MH draws against the Stan draws, where points on the dashed line indicate identical distributions.

Table 4: Comparison of samplers, containing the difference in estimates, the pooled variance and p-value of the two-sample t-test in addition to the Kolomogorov-Smirnov distance between the draws.

	Mean diff.	Pooled var.	p-value	2.5% diff.	Median diff.	97.5% diff.	D
μ	-0.119	90.764	0.304	-0.175	-0.136	0.142	0.010
σ^2	0.288	7703.986	0.769	-1.553	-0.323	6.150	0.009
τ	0.259	367.647	0.248	0.767	0.246	0.243	0.008

So the estimates appear to not differ any more than would be expected by monte-carlo error. The samplers do differ in efficiency, however. The effective sample size of the MH algorithm is much smaller than the effective sample size utilized by stan (due to the covariance between μ and τ). This is due to how each samplers explore the posterior surface; NUTS uses gradient information to move along this ridge, whereas the random walk, with a fixed proposal covariance, takes small steps to keep its acceptance rate. The MH sampler needs no gradients, but NUTS gives the same answer from far fewer draws.

Question f

We interpret the Type 3 posterior biologically using the pooled Metropolis-Hastings draws. Besides the model parameters, we compute the midpoint $\mu + \tau/2$ and the completion date $\mu + \tau$ of moult for each draw, so that their uncertainty accounts for the correlation between μ and τ. Day 1 is 1 July 1978, so day numbers are converted to calendar dates in Table 5.

Table 5: Posterior medians and 95% credible intervals of the biological quantities (Type 3 model). Day 1 is 1 July 1978.

Quantity	Median	95% CrI	Calendar date (95% CrI)
Mean start date μ	129.6	(114.8, 151.5)	7 Nov (23 Oct to 29 Nov)
Mean midpoint $\mu + \tau/2$	176.6	(167.3, 189.4)	24 Dec (14 Dec to 5 Jan)
Mean completion date $\mu + \tau$	223.3	(200.4, 248.3)	8 Feb (16 Jan to 5 Mar)
Duration τ (days)	93.5	(55.9, 128.6)	
SD of start dates σ (days)	17.4	(13.9, 23.1)	
Moult rate (% per day)	1.07	(0.78, 1.79)	

The posterior medians are $\mu = 129.6$, $\tau = 93.5$ and $\sigma = 17.4$ days (Table 5). The population's mean start date of moult is therefore about 7 Nov (95% credible interval 23 Oct to 29 Nov). A bird takes about 94 days, or 3.1 months, to complete its moult, so its moult index rises by about 1.1% per day (although this estimate is imprecise). The mean midpoint of moult, $\mu + \tau/2$, is about 24 Dec and the mean completion date, $\mu + \tau$, is about 8 Feb; the anti-correlation between μ and τ means that we can make more precise estimates on the midpoints and completion dates than implied by independence.

The standard deviation σ describes how synchronised the birds are. About 95% of birds start moulting within $\pm$ 34 days of the mean, that is, between about 3 Oct and 11 Dec. This spread of start dates is comparable to the duration of an individual's moult, so the moults of different birds overlap substantially. Thus, while each bird is in moult for about 3 months, the central 95% of the population is in moult at some point over about 5.3 months, from early October to March.

These estimates should be interpreted with some care. The interval for μ is wide because the Type 3 model uses only the 66 moulting birds, 55 of which were caught on a single day, so the data cannot easily separate an early start with a long moult from a late start with a short moult. The posteriors of μ and τ are also cut off at the prior bounds of 160 and 50, respectively, so these estimates partly reflect the priors. Finally, the estimates describe one season at one site, and assume that all birds moult at the same constant rate and that start dates are normally distributed.

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Appendix A: Question b derivations

Type 2 likelihood and posterior

The likelihood contributions of birds that have already moulted and birds that have not yet started to moult are the same as in the Type 1 model. The moulting birds contribute the density in Equation 10. Thus, the Type 2 likelihood is

$$L(\boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \mu, \sigma^2, \tau) = \prod_{i \in \mathcal{F}} R(t_i) \prod_{i \in \mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i \in \mathcal{N}} P(t_i), \tag{20}$$

and the joint posterior distribution is

$$\pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto \prod_{i \in \mathcal{F}} R(t_i) \prod_{i \in \mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i \in \mathcal{N}} P(t_i)\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau), \tag{21}$$

with the prior distributions in Equation 11.

Data augmentation

The probabilities $R(t_i)$ and $P(t_i)$ in Equation 21 are integrals over the unobserved start date

$$\begin{aligned} R(t_i) &= Pr(T_i \le t_i - \tau) = \int_{-\infty}^{\infty} f_T(T_i)\, I_{\{T_i \le t_i - \tau\}}\, dT_i, \\ P(t_i) &= Pr(T_i > t_i) = \int_{-\infty}^{\infty} f_T(T_i)\, I_{\{T_i > t_i\}}\, dT_i. \end{aligned} \tag{22}$$

The status is fully determined by the start date (T_i), so each integrand in Equation 22 is the joint density of the start date and the bird's status

$$\begin{aligned} p(T_i, s_i = 1 \mid \mu, \sigma^2, \tau) &= f_T(T_i)\, I_{\{T_i \le t_i - \tau\}}, \\ p(T_i, s_i = 3 \mid \mu, \sigma^2) &= f_T(T_i)\, I_{\{T_i > t_i\}}. \end{aligned} \tag{23}$$

T_i for $i \in \mathcal{F} \cup \mathcal{N}$ is treated as a latent variable and samples are drawn from the augmented posterior

$$\begin{aligned} \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}) \propto & \prod_{i \in \mathcal{F}} f_T(T_i)\, I_{\{T_i \le t_i - \tau\}} \prod_{i \in \mathcal{M}} \tau f_T(t_i - y_i\tau) \\ & \times \prod_{i \in \mathcal{N}} f_T(T_i)\, I_{\{T_i > t_i\}}\, \pi(\mu)\, \pi(\sigma^2)\, \pi(\tau). \end{aligned} \tag{24}$$

Integrating out each T_i in Equation 24 returns $R(t_i)$ or $P(t_i)$, so that

$$\int \pi(\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})}, \mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t})\, d\boldsymbol{T}_{(\mathcal{F}\cup\mathcal{N})} = \pi(\mu, \sigma^2, \tau \mid \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}). \tag{25}$$

The posterior distribution of interest, Equation 21, is the marginal distribution of the augmented posterior. Thus, the draws of μ, σ^2 and τ from a sampler targeting the augmented posterior are draws from the required posterior. No latent variables are needed for the moulting birds, since $T_i = t_i - y_i\tau$ is a deterministic function of τ for $i \in \mathcal{M}$. Since $0 < y_i < 1$, the condition $t_i - \tau \leq T_i \leq t_i$ is always satisfied, so no indicator is required for these birds.

The full conditional distribution of each variable is obtained from Equation 24 by treating all other variables as fixed. Thus, any factor that does not involve the variable being updated is absorbed into the proportionality constant.

Conditional distribution of T_i

The conditional distributions of the latent start dates are the same truncated normal distributions as in the Type 1 Gibbs sampler

$$\begin{aligned} T_i \mid \mu, \sigma^2, \tau, s_i = 1, &\quad t_i \sim \mathrm{N}(\mu, \sigma^2)\, I_{\{T_i \leq t_i - \tau\}}, &\quad i \in \mathcal{F}, \\ T_i \mid \mu, \sigma^2, \tau, s_i = 3, &\quad t_i \sim \mathrm{N}(\mu, \sigma^2)\, I_{\{T_i > t_i\}}, &\quad i \in \mathcal{N}, \end{aligned}$$

as given in Equation 13.

Conditional distribution of μ

From Equation 24, with $T_i = t_i - y_i\tau$ for the moulting birds,

$$\begin{aligned} \pi(\mu \mid \sigma^2, \tau, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T}) &\propto \prod_{i \in \mathcal{F}} f_T(T_i) \prod_{i \in \mathcal{M}} f_T(t_i - y_i\tau) \prod_{i \in \mathcal{N}} f_T(T_i)\, \pi(\mu) \\ &= \prod_{i \in \mathcal{F}} f_T(T_i) \prod_{i \in \mathcal{M}} f_T(T_i) \prod_{i \in \mathcal{N}} f_T(T_i)\, \pi(\mu) \\ &= \prod_{i=1}^{n} f_T(T_i)\, \pi(\mu) \\ &\propto \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(T_i - \mu)^2}{2\sigma^2}\right) I_{\{100 \leq \mu \leq 160\}} \\ &\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (T_i - \mu)^2\right) I_{\{100 \leq \mu \leq 160\}}. \end{aligned}$$

Expanding the sum of squares gives

$$\begin{aligned}\sum_{i=1}^{n}(T_i-\mu)^2 &= \sum_{i=1}^{n}\left[(T_i-\bar{T})+(\bar{T}-\mu)\right]^2 \\ &= \sum_{i=1}^{n}(T_i-\bar{T})^2 + 2(\bar{T}-\mu)\sum_{i=1}^{n}(T_i-\bar{T}) + n(\bar{T}-\mu)^2 \\ &= \sum_{i=1}^{n}(T_i-\bar{T})^2 + n(\bar{T}-\mu)^2,\end{aligned}$$

since $\sum_{i=1}^{n}(T_i-\bar{T}) = 0$. Absorbing $\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(T_i-\bar{T})^2\right)$ into the proportionality constant gives

$$\pi(\mu \mid \sigma^2, \tau, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T}) \propto \exp\left(-\frac{n}{2\sigma^2}(\mu-\bar{T})^2\right) I_{\{100\leq\mu\leq160\}},$$

which is the kernel of the truncated normal distribution in Equation 14.

Conditional distribution of σ^2

From Equation 24, with $T_i = t_i - y_i\tau$ for the moulting birds and the IG$(1,1)$ prior $\pi(\sigma^2) \propto (\sigma^2)^{-2}e^{-1/\sigma^2}$,

$$\begin{aligned}\pi(\sigma^2 \mid \mu, \tau, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T}) &\propto \prod_{i\in\mathcal{F}} f_T(T_i) \prod_{i\in\mathcal{M}} f_T(t_i - y_i\tau) \prod_{i\in\mathcal{N}} f_T(T_i)\,\pi(\sigma^2) \\ &= \prod_{i=1}^{n} f_T(T_i)\,\pi(\sigma^2) \\ &\propto (\sigma^2)^{-n/2}\exp\left(-\frac{S}{2\sigma^2}\right)(\sigma^2)^{-2}\exp\left(-\frac{1}{\sigma^2}\right) \\ &= (\sigma^2)^{-(\frac{n}{2}+1)-1}\exp\left(-\frac{1+\frac{S}{2}}{\sigma^2}\right),\end{aligned}$$

which is the kernel of the inverse-gamma distribution in Equation 15.

Conditional distribution of τ

From Equation 24, with $T_i = t_i - y_i\tau$ for the moulting birds

$$\begin{aligned}\pi(\tau \mid \mu, \sigma^2, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T}) &\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} I_{\{T_i\leq t_i-\tau\}}\,\pi(\tau) \\ &\propto \prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \prod_{i\in\mathcal{F}} I_{\{T_i\leq t_i-\tau\}}\, I_{\{50\leq\tau\leq180\}}.\end{aligned}$$

The moulting birds contribute

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\frac{1}{2\sigma^2}\sum_{i\in\mathcal{M}}(t_i - \mu - y_i\tau)^2\right).$$

Expanding the square gives

$$\sum_{i\in\mathcal{M}}(t_i - \mu - y_i\tau)^2 = \sum_{i\in\mathcal{M}}(t_i-\mu)^2 - 2\tau\sum_{i\in\mathcal{M}} y_i(t_i-\mu) + \tau^2\sum_{i\in\mathcal{M}} y_i^2.$$

Absorbing the term in $\sum_{i\in\mathcal{M}}(t_i-\mu)^2$ into the proportionality constant and using β and γ from Equation 12 gives

$$\prod_{i\in\mathcal{M}} \tau f_T(t_i - y_i\tau) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right).$$

Each indicator of a finished bird can be rearranged as $T_i \le t_i - \tau \iff \tau \le t_i - T_i$. The product of the indicators equals one only if this holds for every finished bird, so

$$\prod_{i\in\mathcal{F}} I_{\{T_i \le t_i - \tau\}} = I_{\{\tau \le \min_{i\in\mathcal{F}}(t_i - T_i)\}} = I_{\{\tau \le h\}}.$$

Thus, the duration of moult cannot exceed the time between the start date and the capture date of any finished bird. Combining these terms with the prior gives

$$\pi(\tau \mid \mu, \sigma^2, \boldsymbol{s}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{T}) \propto \tau^m \exp\left(-\beta\tau^2 + \gamma\tau\right) I_{\{50\le\tau\le\min(180,h)\}}.$$

This is the kernel of the truncated MHN distribution in Equation 16, with $\alpha = m + 1$.

Data augmentation for τ

For the `sanderlings` data, $\gamma > 0$ at every iteration, so only the Poisson augmentation is required. Multiplying the conditional distribution of τ by the Poisson probability of $W \mid \tau$ gives the joint density

$$p(\tau, W) \propto \tau^m e^{-\beta\tau^2+\gamma\tau} \cdot \frac{e^{-\gamma\tau}(\gamma\tau)^W}{W!} I_{\{50\le\tau\le\min(180,h)\}} \propto \frac{\gamma^W}{W!}\tau^{m+W}e^{-\beta\tau^2} I_{\{50\le\tau\le\min(180,h)\}},$$

so the term $e^{\gamma\tau}$ cancels. Summing over $W = 0, 1, 2, \ldots$ returns the conditional distribution of τ. The Poisson probabilities sum to one, so W is a valid latent variable. Changing variables from τ to τ^2, with $d\tau = d(\tau^2)/(2\tau)$,

$$\tau^{m+W}e^{-\beta\tau^2}\, d\tau \propto (\tau^2)^{\frac{m+1+W}{2}-1} e^{-\beta\tau^2}\, d(\tau^2),$$

which is the kernel of a Gamma $\left(\frac{m+1+W}{2}, \beta\right)$ distribution for τ^2. Since $\tau > 0$, the truncation $50 \le \tau \le \min(180, h)$ is equivalent to $50^2 \le \tau^2 \le \min(180, h)^2$, which gives Equation 17.

W is drawn from its conditional distribution given the current τ, followed by τ^2 given W. This two-step update leaves the conditional distribution of τ in Equation 16 invariant. If $\gamma < 0$, Lemma 5 of Sun et al. (2021) instead uses a generalised inverse Gaussian latent variable, which is not needed here.

Appendix B: Additional figures

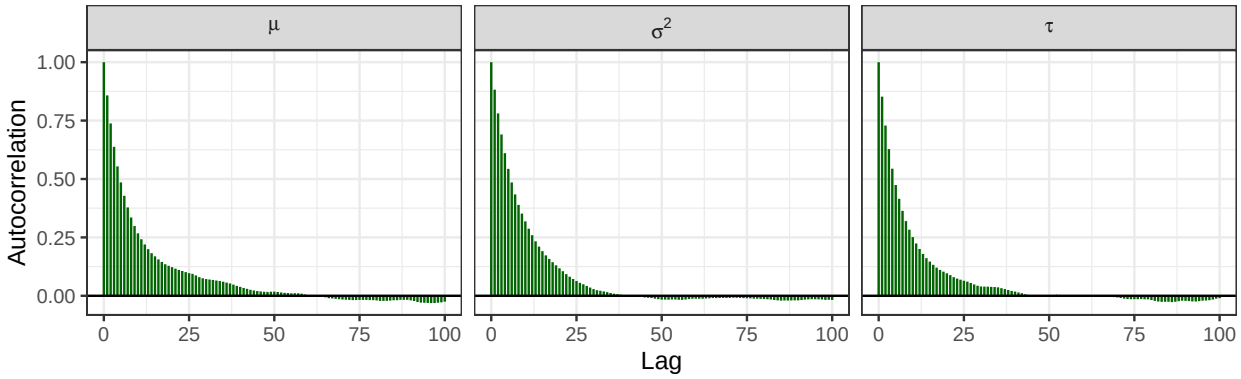


Figure 3: Autocorrelation of the retained draws of the first Metropolis-Hastings chain.

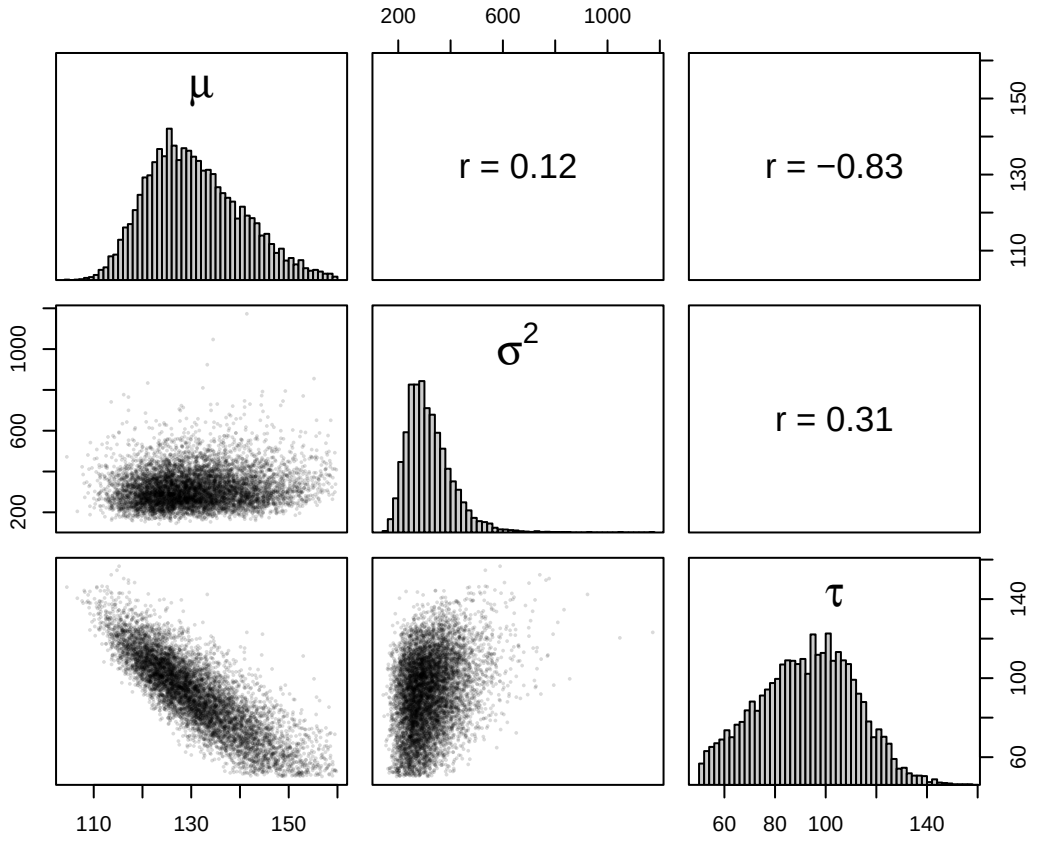


Figure 4: Marginal (diagonal) and pairwise joint (lower panels) posterior distributions with posterior correlations (upper panels), from the pooled Metropolis-Hastings chains (every 20th draw shown).

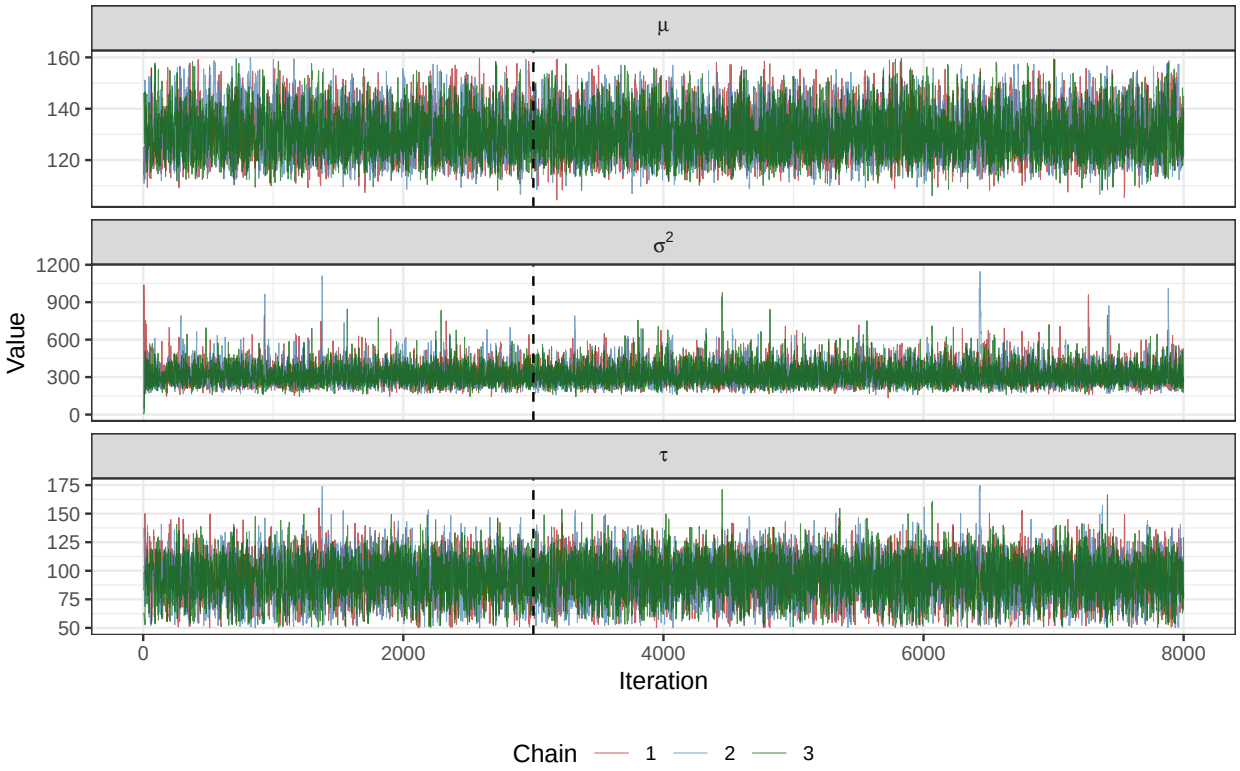


Figure 5: Trace plots of the three Stan chains for the first 8 000 iterations. The end of the warmup is indicated by the dashed line.

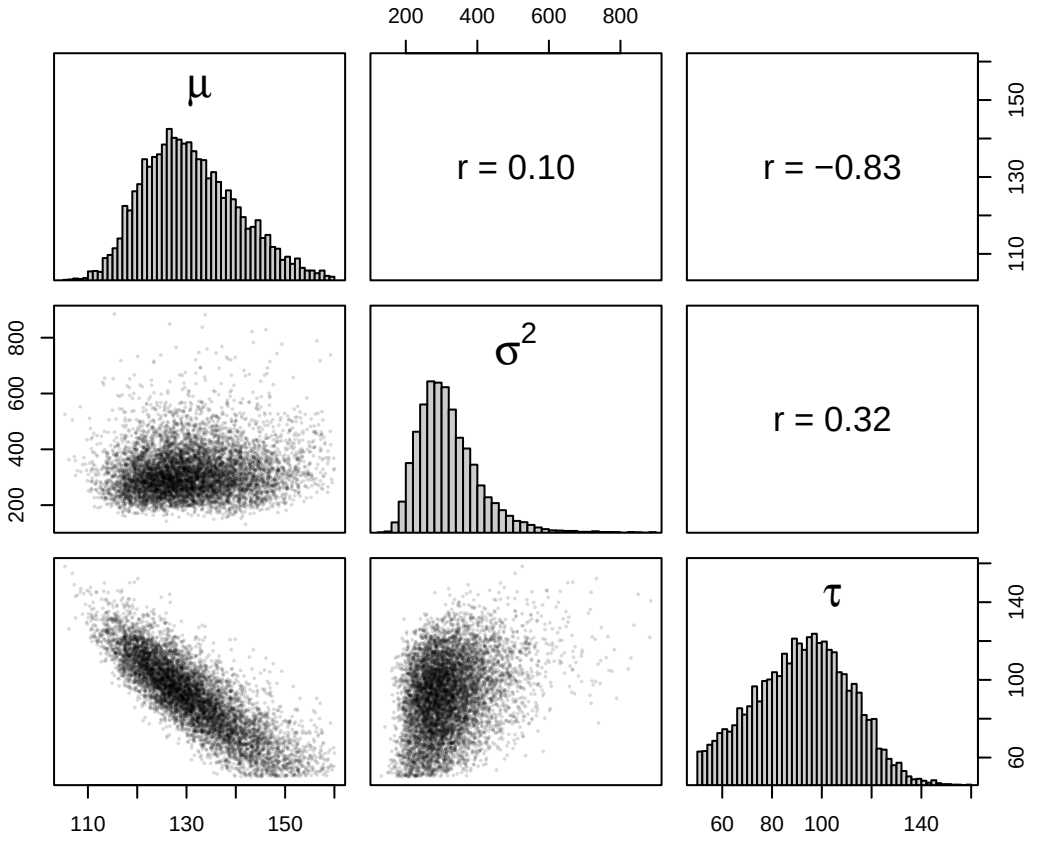


Figure 6: Marginal (diagonal) and pairwise joint (lower panels) posterior distributions with posterior correlations (upper panels), from the pooled Stan chains (every 20th draw shown).

Appendix C: R code

```r
library(mvtnorm)
library(moult)
library(coda)
library(ggplot2)
library(knitr)
library(rstan)
rstan_options(auto_write = TRUE)

# Data and classification scheme:
data(sanderlings)
dat <- sanderlings
dat$status <- ifelse(dat$MIndex == 0, 3, ifelse(dat$MIndex == 1, 1, 2))
moulting <- subset(dat, status == 2)

# The Type 3 model uses only the moulting birds:
y_obs <- moulting$MIndex
t_obs <- moulting$Day

# Names used in Questions d-f:
day <- t_obs
y   <- y_obs
m   <- length(y_obs)
## ---------------------------------------------------------------
## 1. Type 3 log-likelihood (sigma2 is a variance)
## ---------------------------------------------------------------
log_diff_exp <- function(la, lb) la + log1p(-exp(lb - la))

Type_3_Log_Likelihood <- function(y, t, mu, sigma2, tau) {
  sd <- sqrt(sigma2)
  b  <- (t - mu) / sd
  a  <- (t - tau - mu) / sd

  logQ <- ifelse(
    a > 0,
    log_diff_exp(pnorm(a, lower.tail = FALSE, log.p = TRUE),
                 pnorm(b, lower.tail = FALSE, log.p = TRUE)),
    log_diff_exp(pnorm(b, log.p = TRUE),
                 pnorm(a, log.p = TRUE))
  )

  out <- sum(log(tau) + dnorm(t - y * tau, mu, sd, log = TRUE) - logQ)
  if (!is.finite(out)) -Inf else out
}

## ---------------------------------------------------------------
## 2. Parameter transforms: (mu, sigma2, tau) <-> real line
## ---------------------------------------------------------------
to_real <- function(theta) {
  c(qlogis((theta[1] - 100) / 60),
    log(theta[2]),
    qlogis((theta[3] - 50) / 130))
}
```

```r

from_real <- function(phi) {
  c(100 + 60 * plogis(phi[1]),
    exp(phi[2]),
    50 + 130 * plogis(phi[3]))
}

## ------------------------------------------------------------------
## 3. Log posterior on the transformed scale (priors + Jacobians)
## ------------------------------------------------------------------
log_target <- function(phi, y, t) {
  theta <- from_real(phi)
  ll <- Type_3_Log_Likelihood(y, t, theta[1], theta[2], theta[3])
  if (!is.finite(ll)) return(-Inf)

  # Uniform priors x logit Jacobians (constants dropped)
  lp_mu  <- plogis(phi[1], log.p = TRUE) + plogis(-phi[1], log.p = TRUE)
  lp_tau <- plogis(phi[3], log.p = TRUE) + plogis(-phi[3], log.p = TRUE)
  # IG(1,1) prior on sigma2 times Jacobian sigma2
  lp_s2  <- -phi[2] - exp(-phi[2])

  ll + lp_mu + lp_s2 + lp_tau
}

## ------------------------------------------------------------------
## 4. Core MH loop: one step per row of `jitters` (symmetric proposals)
## ------------------------------------------------------------------
mh_run <- function(phi, jitters, y, t) {
  n   <- nrow(jitters)
  cur <- log_target(phi, y, t)
  if (!is.finite(cur)) stop("Starting value has zero density; redraw init.")

  phi_draws <- matrix(NA_real_, n, 3)
  accepted  <- logical(n)

  for (i in seq_len(n)) {
    prop      <- phi + jitters[i, ]
    prop_val  <- log_target(prop, y, t)
    log_alpha <- if (is.finite(prop_val)) min(0, prop_val - cur) else -Inf

    if (log(runif(1)) < log_alpha) {
      phi <- prop
      cur <- prop_val
      accepted[i] <- TRUE
    }
    phi_draws[i, ] <- phi
  }
  list(phi_draws = phi_draws, last = phi, accept_rate = mean(accepted))
}

## ------------------------------------------------------------------
## 5. Two-phase sampler
##    Phase 1: burn-in with isotropic jitter -> covariance of phi draws
##    Phase 2: all iterations retained, proposals ~ N(0, scale * Sigma)
```

```r
## ------------------------------------------------------------------
MH_Type_3 <- function(y, t, init, iterations = 50000,
                      burn = 3000, burn_sd = 0.5,
                      cov_frac = 0.5,
                      scale = 2.38^2 / 3) {

  phi0 <- to_real(init)

  # Phase 1
  burn_jit <- rmvnorm(burn, mean = rep(0, 3), sigma = burn_sd^2 * diag(3))
  b <- mh_run(phi0, burn_jit, y, t)

  keep  <- seq.int(floor(burn * (1 - cov_frac)) + 1, burn)
  Sigma <- cov(b$phi_draws[keep, , drop = FALSE])
  if (any(!is.finite(Sigma)) || det(Sigma) < 1e-12)
    stop("Burn-in covariance is degenerate (chain barely moved?). Reduce
    burn_sd and retry.")

  # Phase 2
  main_jit <- rmvnorm(iterations, mean = rep(0, 3), sigma = scale * Sigma)
  m <- mh_run(b$last, main_jit, y, t)

  draws <- t(apply(m$phi_draws, 1, from_real))
  colnames(draws) <- c("mu", "sigma2", "tau")
  burn_draws <- t(apply(b$phi_draws, 1, from_real))
  colnames(burn_draws) <- c("mu", "sigma2", "tau")

  list(draws       = draws,          # original scale, retained
       burn_draws  = burn_draws,     # original scale, burn-in (for trace
    plots)
       phi_draws   = m$phi_draws,    # transformed scale
       Sigma       = Sigma,          # tuned covariance (transformed space)
       burn_accept = b$accept_rate,
       accept_rate = m$accept_rate)
}

## ------------------------------------------------------------------
## 6. Random starting values from the prior ranges (redraw if bad)
## ------------------------------------------------------------------
random_init <- function(y, t) {
  repeat {
    init <- c(runif(1, 100, 160), 1 / rgamma(1, 1, 1), runif(1, 50, 180))
    if (is.finite(log_target(to_real(init), y, t))) return(init)
  }
}
## ------------------------------------------------------------------
## 1. Run three chains from random (prior-based) starting values
## ------------------------------------------------------------------
set.seed(2026)
n_chains <- 3
n_iter   <- 50000

# Retry with a fresh start if the burn-in covariance comes out degenerate
run_chain <- function() {
```

```
  repeat {
    init <- random_init(y_obs, t_obs)
    fit  <- tryCatch(MH_Type_3(y_obs, t_obs, init = init, iterations = n_iter),
                     error = function(e) NULL)
    if (!is.null(fit)) { fit$init <- init; return(fit) }
  }
}
fits <- lapply(seq_len(n_chains), function(i) run_chain())
pars <- c("mu", "sigma2", "tau")
burn <- 3000

## Retained draws as coda objects, with sigma = sqrt(sigma2) added
add_sigma <- function(d) cbind(d, sigma = sqrt(d[, "sigma2"]))
chains <- mcmc.list(lapply(fits, function(f) mcmc(add_sigma(f$draws))))

## Convergence diagnostics and summaries
rhat      <- gelman.diag(chains, multivariate = FALSE)$psrf[, "Point est."]
ess       <- effectiveSize(chains)
pooled    <- as.matrix(chains)
cor_mat   <- cor(pooled[, pars])
acc_rates <- sapply(fits, `[[`, "accept_rate")
burn_acc  <- sapply(fits, `[[`, "burn_accept")

## Names used in Questions e-f
mh_post <- chains
mh_mat  <- pooled
chain_tab <- data.frame(
  Chain = seq_len(n_chains),
  mu0   = sapply(fits, function(f) f$init[1]),
  s20   = sapply(fits, function(f) f$init[2]),
  tau0  = sapply(fits, function(f) f$init[3]),
  burn  = burn_acc,
  main  = acc_rates
)
colnames(chain_tab) <- c("Chain", "$\\mu^{(0)}$", "$\\sigma^{2(0)}$", "$\\tau^{(0)}$",
                         "Burn-in acceptance", "Acceptance rate")
kable(chain_tab, digits = 2, escape = FALSE)
n_show <- 8000
p_lab  <- c(mu = "mu", sigma2 = "sigma^2", tau = "tau")

trace_df <- do.call(rbind, lapply(seq_len(n_chains), function(j) {
  th <- rbind(fits[[j]]$burn_draws, fits[[j]]$draws)[1:n_show, ]
  data.frame(iteration = rep(1:n_show, 3),
             value     = c(th[, "mu"], th[, "sigma2"], th[, "tau"]),
             parameter = factor(rep(p_lab, each = n_show), levels = p_lab),
             chain     = factor(j))
}))

ggplot(trace_df, mapping = aes(x = iteration, y = value, colour = chain)) +
  geom_line(linewidth = 0.2, alpha = 0.7) +
  geom_vline(xintercept = burn, colour = "black", linetype = "dashed") +
  facet_wrap(~ parameter, scales = "free_y", ncol = 1, labeller = label_parsed
```

```
    ) +
  scale_colour_manual(values = c("firebrick", "steelblue", "darkgreen")) +
  labs(x = "Iteration", y = "Value", colour = "Chain") +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom")
mh_tab <- data.frame(
  Mean   = colMeans(pooled),
  SD     = apply(pooled, 2, sd),
  Q2.5   = apply(pooled, 2, quantile, 0.025),
  Median = apply(pooled, 2, quantile, 0.5),
  Q97.5  = apply(pooled, 2, quantile, 0.975),
  ESS    = round(ess[colnames(pooled)]),
  Rhat   = sprintf("%.2f", rhat[colnames(pooled)])
)
colnames(mh_tab) <- c("Mean", "SD", "2.5%", "Median", "97.5%", "ESS", "$\\hat{
    R}$")
rownames(mh_tab) <- c("$\\mu$", "$\\sigma^2$", "$\\tau$", "$\\sigma$")
kable(mh_tab, digits = 2, escape = FALSE, align = "r")
# Type 3 model written in Stan:
modelString = "
  data {
    int<lower=1> m;                    // number of moulting birds
    vector[m] t;                       // capture day of each bird
    vector<lower=0, upper=1>[m] y;     // moult index
  }

  parameters {
    real<lower=100, upper=160> mu;     // mean start date of moult
    real<lower=0> sigma2;              // variance of start dates
    real<lower=50, upper=180> tau;     // duration of moult
  }

  transformed parameters {
    real<lower=0> sigma = sqrt(sigma2);
  }

  model {
    // Priors:
    mu     ~ uniform(100, 160);
    sigma2 ~ inv_gamma(1, 1);
    tau    ~ uniform(50, 180);

    // Likelihood:
    for (i in 1:m) {
      // log Q(t_i), computed stably with log_diff_exp
      real logQ = log_diff_exp(normal_lcdf(t[i] | mu, sigma),
                               normal_lcdf(t[i] - tau | mu, sigma));
      target += log(tau) + normal_lpdf(t[i] - y[i] * tau | mu, sigma) - logQ;
    }
  }
"
library(rstan)
rstan_options(auto_write = TRUE)

```

```r
# Translate the model to C++ and compile:
stan_mod = stan_model(model_code = modelString)

# Specify data:
stan_data = list(
  m = m,
  t = day,
  y = y)

# Generate posterior sample:
stanFit = sampling(stan_mod,
                   data = stan_data,
                   chains = 3,
                   iter = 53000,
                   warmup = 3000,
                   thin = 1,
                   seed = 2026)
stan_sum = summary(stanFit, pars = c("mu", "sigma2", "tau", "sigma"),
                   probs = c(0.025, 0.5, 0.975))$summary

n_div     = sum(get_divergent_iterations(stanFit))       # divergent
    transitions
n_td      = sum(get_max_treedepth_iterations(stanFit))   # max tree depth hits
stan_rhat = max(stan_sum[, "Rhat"])
stan_ess  = stan_sum[c("mu", "sigma2", "tau"), "n_eff"]
stan_time = rowSums(get_elapsed_time(stanFit))           # seconds per chain
stan_tab = data.frame(stan_sum[, c("mean", "sd", "2.5%", "50%", "97.5%", "
    n_eff")])
stan_tab$Rhat = sprintf("%.2f", stan_sum[, "Rhat"])
colnames(stan_tab) = c("Mean", "SD", "2.5%", "Median", "97.5%", "ESS", "$\\hat
    {R}$")
rownames(stan_tab) = c("$\\mu$", "$\\sigma^2$", "$\\tau$", "$\\sigma$")
stan_tab$ESS = round(stan_tab$ESS)

knitr::kable(stan_tab, digits = 2, escape = FALSE, align = "r")
pars4     = c("mu", "sigma2", "tau", "sigma")
pars3     = pars4[1:3]
p_lab     = c(mu = "mu", sigma2 = "sigma^2", tau = "tau")
mh_d      = as.data.frame(mh_mat[, pars4])
stan_d    = as.data.frame(as.matrix(stanFit, pars = pars4))
stan_ess4 = stan_sum[pars4, "n_eff"]
p_expr = list(expression(mu), expression(sigma^2), expression(tau))
pr     = seq(0.005, 0.995, by = 0.005)
op     = par(mfrow = c(2, 3), mar = c(4, 4, 2, 1), mgp = c(2.4, 0.8, 0))

## Top row: overlaid marginal posterior densities
for (k in 1:3) {
  p  = pars3[k]
  dm = density(mh_d[[p]])
  ds = density(stan_d[[p]])
  plot(dm, col = "purple", lwd = 1.5, main = p_expr[[k]], xlab = "Value",
       ylab = "Posterior density", ylim = c(0, max(dm$y, ds$y)))
  lines(ds, col = "steelblue", lwd = 1.5, lty = 2)
  if (k == 1) {
```

```
    legend("topright", legend = c("MH", "Stan"), col = c("purple", "steelblue"),
           lty = c(1, 2), lwd = 1.5, bty = "n", cex = 0.8)
  }
}

## Bottom row: quantile-quantile plots
for (k in 1:3) {
  p = pars3[k]
  plot(quantile(stan_d[[p]], pr), quantile(mh_d[[p]], pr), pch = 16, cex = 0.5,
       main = p_expr[[k]], xlab = "Stan quantiles", ylab = "MH quantiles")
  abline(0, 1, col = "red", lty = 2)
}
par(op)
use_ess = TRUE    # TRUE: effective sample sizes in the standard error; FALSE: naive, all draws independent
qprobs  = c(0.025, 0.5, 0.975)

cmp = do.call(rbind, lapply(pars3, function(p) {
  x   = mh_d[[p]]
  s   = stan_d[[p]]
  n1  = length(x)
  n2  = length(s)
  ne1 = if (use_ess) ess[p]       else n1
  ne2 = if (use_ess) stan_ess4[p] else n2
  sp2 = ((n1 - 1) * var(x) + (n2 - 1) * var(s)) / (n1 + n2 - 2)   # pooled variance
  d   = mean(x) - mean(s)
  tt  = d / sqrt(sp2 * (1 / ne1 + 1 / ne2))
  pv  = 2 * pt(-abs(tt), df = ne1 + ne2 - 2)
  qd  = unname(quantile(x, qprobs) - quantile(s, qprobs))
  data.frame(diff_mean = d, diff_sd = d / sqrt(sp2), pooled_var = sp2,
             t = unname(tt), p = unname(pv),
             d_lo = qd[1], d_med = qd[2], d_hi = qd[3],
             D = unname(suppressWarnings(ks.test(x, s)$statistic)))
}))
rownames(cmp) = pars3

cmp_tab = cmp[, c("diff_mean", "pooled_var", "p", "d_lo", "d_med", "d_hi", "D")]
colnames(cmp_tab) = c("Mean diff.", "Pooled var.", "$p$-value",
                      "2.5% diff.", "Median diff.", "97.5% diff.", "$D$")
rownames(cmp_tab) = c("$\\mu$", "$\\sigma^2$", "$\\tau$")
kable(cmp_tab, digits = 3, escape = FALSE)
eff = data.frame(ess_mh   = unname(ess[pars4]),
                 ess_stan = unname(stan_ess4))
eff$ratio     = eff$ess_stan / eff$ess_mh
eff$frac_mh   = eff$ess_mh   / nrow(mh_d)
eff$frac_stan = eff$ess_stan / nrow(stan_d)
# Day 1 = 1 July 1978 (sanderlings help file):
to_date = function(d) sub("^0", "", format(as.Date("1978-06-30") + round(d), "%d %b"))

```

```r
post       = as.data.frame(mh_mat[, c("mu", "sigma2", "tau")])  # use stan_d for the Stan draws
post$sigma = sqrt(post$sigma2)
post$mid   = post$mu + post$tau / 2      # midpoint of moult
post$end   = post$mu + post$tau          # completion date
post$rate  = 100 / post$tau              # % of moult completed per day

qs = t(sapply(post[, c("mu", "mid", "end", "tau", "sigma", "rate")],
              quantile, probs = c(0.025, 0.5, 0.975)))

# Central 95% of individual start dates, and the span of the whole population's moult:
lo95     = qs["mu", "50%"] - 1.96 * qs["sigma", "50%"]
hi95     = qs["mu", "50%"] + 1.96 * qs["sigma", "50%"]
pop_span = (hi95 + qs["tau", "50%"]) - lo95
dg       = c(1, 1, 1, 1, 1, 2)
fmt      = function(x) unname(mapply(function(v, d) formatC(v, format = "f", digits = d), x, dg))
is_date = c(TRUE, TRUE, TRUE, FALSE, FALSE, FALSE)

bio = data.frame(
  Quantity = c("Mean start date $\\mu$", "Mean midpoint $\\mu + \\tau/2$",
               "Mean completion date $\\mu + \\tau$", "Duration $\\tau$ (days)",
               "SD of start dates $\\sigma$ (days)", "Moult rate (\\% per day)"),
  Median   = fmt(qs[, "50%"]),
  CrI      = sprintf("(%s, %s)", fmt(qs[, "2.5%"]), fmt(qs[, "97.5%"])),
  Date     = ifelse(is_date,
                    sprintf("%s (%s to %s)", to_date(qs[, "50%"]),
                            to_date(qs[, "2.5%"]), to_date(qs[, "97.5%"])), "")
)
colnames(bio) = c("Quantity", "Median", "95% CrI", "Calendar date (95% CrI)")
rownames(bio) = NULL
kable(bio, escape = FALSE, align = "lrrr")
acf_df <- do.call(rbind, lapply(names(p_lab), function(p) {
  a <- acf(as.numeric(chains[[1]][, p]), lag.max = 100, plot = FALSE)
  data.frame(lag = as.numeric(a$lag), acf = as.numeric(a$acf),
             parameter = factor(p_lab[p], levels = p_lab))
}))

ggplot(acf_df, mapping = aes(x = lag, y = acf)) +
  geom_segment(aes(xend = lag, yend = 0), colour = "darkgreen") +
  geom_hline(yintercept = 0) +
  facet_wrap(~ parameter, nrow = 1, labeller = label_parsed) +
  labs(x = "Lag", y = "Autocorrelation") +
  theme_bw(base_size = 10)
thin_pool <- pooled[seq(1, nrow(pooled), by = 20), pars]

panel_cor <- function(x, y, ...) {
  usr <- par("usr"); on.exit(par(usr = usr))
  par(usr = c(0, 1, 0, 1))
  text(0.5, 0.5, sprintf("r = %.2f", cor(x, y)), cex = 1.6)
}
```

```r
panel_pts <- function(x, y, ...) points(x, y, pch = 16, cex = 0.4, col = rgb(0, 0, 0, 0.15))
panel_hist <- function(x, ...) {
  usr <- par("usr"); on.exit(par(usr = usr))
  h <- hist(x, plot = FALSE, breaks = 40)
  par(usr = c(usr[1:2], 0, max(h$counts) * 1.5))
  rect(h$breaks[-length(h$breaks)], 0, h$breaks[-1], h$counts, col = "grey80")
}
pairs(thin_pool, lower.panel = panel_pts, upper.panel = panel_cor,
      diag.panel = panel_hist,
      labels = expression(mu, sigma^2, tau))
# Draws including warmup: array [iteration, chain, parameter]
stan_arr <- rstan::extract(stanFit, pars = c("mu", "sigma2", "tau"),
                           permuted = FALSE, inc_warmup = TRUE)
n_show <- 8000
warm   <- 3000
p_lab  <- c(mu = "mu", sigma2 = "sigma^2", tau = "tau")

stan_trace_df <- do.call(rbind, lapply(seq_len(dim(stan_arr)[2]), function(j) {
  data.frame(iteration = rep(1:n_show, 3),
             value     = c(stan_arr[1:n_show, j, "mu"],
                           stan_arr[1:n_show, j, "sigma2"],
                           stan_arr[1:n_show, j, "tau"]),
             parameter = factor(rep(p_lab, each = n_show), levels = p_lab),
             chain     = factor(j))
}))

ggplot(stan_trace_df, mapping = aes(x = iteration, y = value, colour = chain)) +
  geom_line(linewidth = 0.2, alpha = 0.7) +
  geom_vline(xintercept = warm, colour = "black", linetype = "dashed") +
  facet_wrap(~ parameter, scales = "free_y", ncol = 1, labeller = label_parsed) +
  scale_colour_manual(values = c("firebrick", "steelblue", "darkgreen")) +
  labs(x = "Iteration", y = "Value", colour = "Chain") +
  theme_bw(base_size = 10) +
  theme(legend.position = "bottom")
stan_mat  <- as.matrix(stanFit, pars = c("mu", "sigma2", "tau"))
thin_stan <- stan_mat[seq(1, nrow(stan_mat), by = 20), ]

pairs(thin_stan, lower.panel = panel_pts, upper.panel = panel_cor,
      diag.panel = panel_hist,
      labels = expression(mu, sigma^2, tau))
```